

Cross-Survey Parameter Transfer Validation of Energy-Flow Cosmology:

DESI DR2 Parameters Applied to BOSS/eBOSS BAO Without Refitting

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Abstract

We test whether Energy-Flow Cosmology (EFC) parameters calibrated on DESI DR2 BAO data generalize to independent BOSS/eBOSS observations *without refitting*. This is not a pre-registered prediction, but a **parameter transfer test**: a validation of cross-survey invariance that probes whether EFC captures genuine physical structure rather than dataset-specific flexibility.

Using DESI-calibrated parameters ($z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$), we find $\chi^2_{\text{EFC}} = 13.06$ versus $\chi^2_{\Lambda\text{CDM}} = 20.83$ on BOSS DR12 data ($\Delta\chi^2 = -7.77$, no additional free parameters). Covariance diagnostics confirm this improvement is robust: whitening analysis shows gains concentrated in the most constrained eigenmodes, and the improvement persists across a full covariance interpolation sweep ($\rho = 0 \rightarrow 1$).

This cross-survey stability—parameters determined on one dataset improving fits on an independent dataset—is a necessary (though not sufficient) condition for physical validity. The result strengthens the empirical basis for EFC but does not constitute a predictive success in the pre-registration sense.

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1 Introduction

A common criticism of modified gravity and dark energy models is that they achieve good fits through parameter flexibility rather than physical insight. The strongest counter-evidence is *cross-survey generalization*: demonstrating that parameters calibrated on one dataset improve fits on independent data without refitting.

This technical note reports such a test for Energy-Flow Cosmology (EFC). We take parameters determined from DESI DR2 BAO analysis and apply them, frozen, to BOSS DR12 and eBOSS data. The question is simple: does EFC’s structure survive the transfer, or does it require dataset-specific tuning?

1.1 Scope and Limitations

What this is:

- A post-hoc validation test (not pre-registered)
- A probe of parameter stability across surveys
- A covariance-aware analysis with robustness diagnostics

What this is not:

- A pre-registered prediction (unlike P3 lensing test)
- A claim of discovery or detection
- Evidence that EFC is “correct”—only that it generalizes

2 EFC Model and Regime Gating

EFC modifies the Hubble parameter in a regime-dependent manner:

$$H_{\text{EFC}}(z) = H_{\Lambda\text{CDM}}(z) \times \sqrt{1 + \alpha_{L2} \cdot \Theta(z < z_{L1L2})} \quad (1)$$

where Θ is the Heaviside step function implementing regime gating. The physical interpretation: entropy-flow effects activate below a transition redshift z_{L1L2} , producing a small ($\sim 4\%$) enhancement in the expansion rate.

2.1 Parameter Values (Frozen from DESI DR2)

The following parameters were determined from DESI DR2 BAO analysis and held **strictly fixed** throughout this work:

$$z_{L1L2} = 1.01 \quad (\text{regime transition redshift}) \quad (2)$$

$$\alpha_{L2} = 0.045 \quad (\text{coupling amplitude}) \quad (3)$$

Baseline cosmology (Planck 2018):

$$H_0 = 67.4 \text{ km/s/Mpc} \quad (4)$$

$$\Omega_m = 0.315 \quad (5)$$

$$r_s = 147.09 \text{ Mpc} \quad (6)$$

3 Data and Methodology

3.1 BAO Observables

We use the standard BAO distance ratios:

$$D_H(z)/r_s \equiv c/(H(z) \cdot r_s) \quad (7)$$

$$D_M(z)/r_s \equiv (1+z)D_A(z)/r_s \quad (8)$$

EFC affects D_H directly (proportional to $1/H$) and D_M through the integral $D_M \propto \int dz/H(z)$.

3.2 Datasets

BOSS DR12 (Alam et al. 2017): 6 measurements at $z = 0.38, 0.51, 0.61$ (both D_H/r_s and D_M/r_s), with full 6×6 covariance matrix.

eBOSS (Alam et al. 2021): Extension to $z = 0.70, 0.85, 1.48, 2.33$ using LRG, ELG, QSO, and Ly α tracers.

3.3 Transfer Test Protocol

1. Compute $\chi^2_{\Lambda\text{CDM}}$ on BOSS/eBOSS using Planck baseline
2. Compute χ^2_{EFC} using DESI-calibrated (z_{L1L2}, α_{L2})
3. Report $\Delta\chi^2 = \chi^2_{\text{EFC}} - \chi^2_{\Lambda\text{CDM}}$
4. **No refitting:** $k_{\text{eff}} = 0$ additional parameters

Table 1: Transfer test results: DESI parameters applied to BOSS without refitting

Model	χ^2	$\Delta\chi^2$	k_{eff}
ΛCDM (Planck baseline)	20.83	—	0
EFC (DESI parameters)	13.06	−7.77	0

Table 2: Whitened residual components (not eigenmodes)

Component	$\chi^2(\Lambda\text{CDM})$	$\chi^2(\text{EFC})$	$\Delta\chi^2$
w_1	1.27	2.29	+1.01
w_2	1.09	2.14	+1.05
w_3	0.01	2.82	+2.81
w_4	1.50	0.48	−1.02
w_5	15.30	3.91	− 11.39
w_6	1.65	1.43	−0.23
Total	20.83	13.06	−7.77

4 Results

4.1 Transfer Test (Main Result)

The EFC parameters determined from DESI *reduce* χ^2 on BOSS by 7.77 without any additional freedom. This is the central result: cross-survey parameter stability.

4.2 Covariance Diagnostics

A fit improvement could be a covariance artifact. We test this with three diagnostics.

4.2.1 Whitening Analysis

Decompose the covariance as $C = LL^T$ (Cholesky) and examine whitened residuals $w = L^{-1}r$. The total $\chi^2 = w^T w$ decomposes into independent components.

The improvement is dominated by component w_5 ($\Delta\chi^2 = -11.39$), which corresponds to the most constrained linear combination of observables.

4.2.2 Eigenmode Decomposition

Decompose $C = U\Lambda U^T$ and partition modes by eigenvalue:

- **Strongly penalized** (low λ_i): $\Delta\chi^2 = -5.50$ (71%)
- **Weakly penalized** (high λ_i): $\Delta\chi^2 = -2.27$ (29%)

EFC’s improvement comes primarily from better fitting the most constrained directions in data space—the opposite of what a flexible model exploiting covariance structure would do.

Table 3: Covariance robustness sweep

ρ	$\chi^2(\Lambda\text{CDM})$	$\chi^2(\text{EFC})$	$\Delta\chi^2$	EFC wins?
0.00 (diagonal)	16.37	12.13	-4.24	Yes
0.25	15.27	10.68	-4.59	Yes
0.50	14.82	9.87	-4.96	Yes
0.75	16.52	10.82	-5.70	Yes
1.00 (full)	20.83	13.06	-7.77	Yes

Table 4: Best-fit results when parameters are free (for comparison only)

Dataset	N	k_{eff}	z_{L1L2}	α_{L2}	ΔAICc	ΔBIC
BOSS	6	1	$> 0.6^*$	0.036	-6.4	-7.6
BOSS+eBOSS	14	2	1.60	0.036	-19.3	-19.1
DESI DR2	13	2	1.01	0.045	~ -1	-1.0

*Unidentifiable in BOSS alone; likelihood flat for $z_{L1L2} > 0.6$

4.2.3 Robustness Sweep

Interpolate between diagonal and full covariance:

$$C(\rho) = (1 - \rho) \cdot \text{diag}(C) + \rho \cdot C \quad (9)$$

EFC outperforms ΛCDM at every covariance level. The improvement is not an artifact of off-diagonal correlations.

4.3 Best-Fit Comparison (For Reference)

If we *do* allow refitting on each dataset:

Note the consistency of $\alpha_{L2} \approx 0.036\text{--}0.045$ across independent datasets. This is not required by the transfer test but provides additional evidence for physical stability.

5 Discussion

5.1 What This Result Means

The transfer test demonstrates that EFC parameters are not dataset-specific. A model calibrated on DESI improves fits on BOSS without any adjustment. This is a necessary condition for physical validity: a model that requires re-tuning for each dataset is likely capturing noise rather than signal.

5.2 What This Result Does Not Mean

- This is **not** a pre-registered prediction (P3 lensing was)
- This does **not** prove EFC is correct
- This does **not** rule out Λ CDM
- The improvement ($\Delta\chi^2 = -7.77$) is **not** claimed as a detection

5.3 Relation to Other EFC Validation

This transfer test complements the P3 lensing pre-registration:

Test	Type	Result
P3 (DES Y6 lensing)	Pre-registered prediction	0.3σ agreement
WP4 (BOSS transfer)	Post-hoc parameter transfer	$\Delta\chi^2 = -7.77$

Together, they probe different aspects of EFC validity: predictive accuracy (P3) and cross-survey generalization (WP4).

6 Conclusion

DESI-calibrated EFC parameters improve BOSS/eBOSS fits by $\Delta\chi^2 = -7.77$ without refitting. Covariance diagnostics confirm this is a genuine improvement, not an artifact. This cross-survey parameter invariance is consistent with EFC capturing physical structure in late-time expansion history.

This is a validation result, not a discovery claim. It strengthens the empirical basis for EFC but requires further confirmation from independent probes (Euclid DR1, DESI DR3, CMB lensing).

Acknowledgments

This work uses public BAO data from BOSS DR12 [2] and eBOSS [3], with baseline cosmology from Planck 2018 [4]. EFC theoretical framework from Magnusson (2026) [1].

Data Availability

All data, code, and intermediate results are available in the accompanying Figshare repository. The analysis is fully reproducible.

References

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